

Anti-Bias Backtesting Audit Checklist

The 10 critical operational checkpoints separating production alpha from rejected take-home submissions

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⚡ Pillar 1: Temporal Integrity & Look-Ahead Elimination

1. Mandatory Order Execution Lag

Portfolio weights computed at close `t` execute at `t+1`. Always enforce `exec_weights = target_weights.shift(1)`. Evaluating same-day returns with same-day weights causes instant disqualification on institutional desks.

2. Point-in-Time Accounting Data

Quarterly financial statements experience substantial latency between quarter-end and public release. Never index by fiscal period end date; strictly anchor to official public disclosure timestamps (`filing_date`).

3. Cross-Sectional Z-Score Standardization

Never standardize factor scores using full-sample means or standard deviations across time. Normalization must be strictly cross-sectional per date: `df.sub(df.mean(axis=1), axis=0).div(df.std(axis=1), axis=0)`.

💡 Pillar 2: Microstructure Realism & Execution Frictions

4. Non-Linear Slippage & Square-Root Law

Assuming instantaneous fills at midpoint is financial fiction. Institutional market impact scales concavely: $\text{Impact} \approx Y * \text{Daily_Vol} * \sqrt{Q / \text{ADV}}$. Trading illiquid names erodes 100% of theoretical alpha.

5. Turnover Friction & Exchange Fees

High-turnover models must clear brokerage, exchange tariffs, and bid-ask spreads. Deduct a minimum of 15 bps (0.15%) per traded leg in simulation engines.

6. Borrow Rates on Short Legs

Market-neutral and long/short equity models require active stock borrowing. Borrow rates often range from 2% to 15%+ on hard-to-borrow stocks. Omitting borrow costs severely overstates short alpha.

Pillar 3: Statistical Hygiene & p-Hacking Prevention

7. Survivorship Bias Elimination

Never construct historical universes using current index constituents. Delisted, liquidated, and acquired entities must be systematically retained throughout historical windows.

8. Purged & Embargoed K-Fold Cross-Validation

Standard K-Fold is invalid on autocorrelated financial time series due to label overlap. Use Purged K-Fold with an explicit multi-day embargo quarantine between train and test splits.

9. Multiple Testing Deflated Sharpe Ratio (DSR)

If you screened 2,000 parameter permutations, Extreme Value Theory ensures the top Sharpe ratio is pure noise. Deflate observed Sharpe by total trials and return non-normality.

10. Coherent Tail Risk (CVaR / Expected Shortfall)

95% VaR is oblivious to the severity of catastrophic drawdowns beyond the cutoff. Institutional risk committees demand Conditional VaR: $CVaR_{95} = E[R \mid R \leq VaR_{95}]$.

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